

**FIFTH SEMESTER DIPLOMA EXAMINATION IN
ENGINEERING/TECHNOLOGY**

**EMBEDDED SYSTEMS
MODEL QUESTION PAPER**

[Time: 3 Hrs]

(Maximum marks: 100)

PART-A

(Maximum marks: 10)

I Answer all questions in one or two sentences. Each question carries 2 marks

Marks

1. Define an embedded system.
2. Give the use of EQU assembler directive.
3. Name I/O register associated with stack and give its size in bits.
4. Which is the Global Interrupt Enable bit?
5. Name the architecture used in AVR microcontrollers. (5x2=10)

PART-B

(Maximum marks: 30)

II Answer any five of the following questions. Each question carries 6 marks)

1. Explain different flags in status register of AVR.
2. Explain different data types used in AVR C programming.
3. Explain any 4 conditional branching instructions of Atmega32.
4. Explain different types of embedded OS.
5. Compare Subroutines with Macros.
6. Write some applications of embedded systems.
7. Give different steps in executing an interrupt. (5x6=30)

PART-C

(Maximum marks: 60)

(Answer one full question from each unit. Each full question carries 15 marks)

UNIT-I

III. Draw the architecture of ATmega 32. (15)

OR

IV (a) Briefly describe the different classifications of AVR family. (8)

(b) Explain data memory of Atmega 32. (7)

UNIT-II

V (a) Write an Assembly Language Program to generate a square on Port B Pin 5. (7)

(b) Explain the need for initializing stack and give a simple example for that. (5)

(c) Define pipelining. (3)

OR

VI (a) Write an assembly language program to convert the BCD number 67H into ASCII and store the result in SRAM locations 0x75 and 0x76 (7)

(b) Explain rotate and shift instructions with example. (8)

UNIT-III

VII (a) Explain functions of basic registers of Timers in AVR (8)

(b) Write an AVR C- program to toggle all pins of Port B with some delay. (6)

OR

VIII (a) Explain how interrupts enabled/disabled in AVR and specify the use of GICR register (6)

(b) Explain the logical operators use in AVR C with example. (9)

UNIT-IV

IX Explain specialities of Embedded systems. (15)

OR

X Explain software architecture of an Embedded system. (15)